SCEC TPV104, strike slip rate at the nine on-fault stations (rows: depth 3 / 7.5 / 12 km; columns: strike)

x2_-12_x3_3 (strike -12 km, depth 3 km)

x2_0_x3_3 (strike 0 km, depth 3 km)

x2_12_x3_3 (strike 12 km, depth 3 km)

x2_-9_x3_7.5 (strike -9 km, depth 7.5 km)

x2_0_x3_7.5 (strike 0 km, depth 7.5 km)

x2_9_x3_7.5 (strike 9 km, depth 7.5 km)

x2_-12_x3_12 (strike -12 km, depth 12 km)

x2_0_x3_12 (strike 0 km, depth 12 km)

x2_12_x3_12 (strike 12 km, depth 12 km)

strike slip rate (m/s)

time (s)

DRDG3D 200 m O4

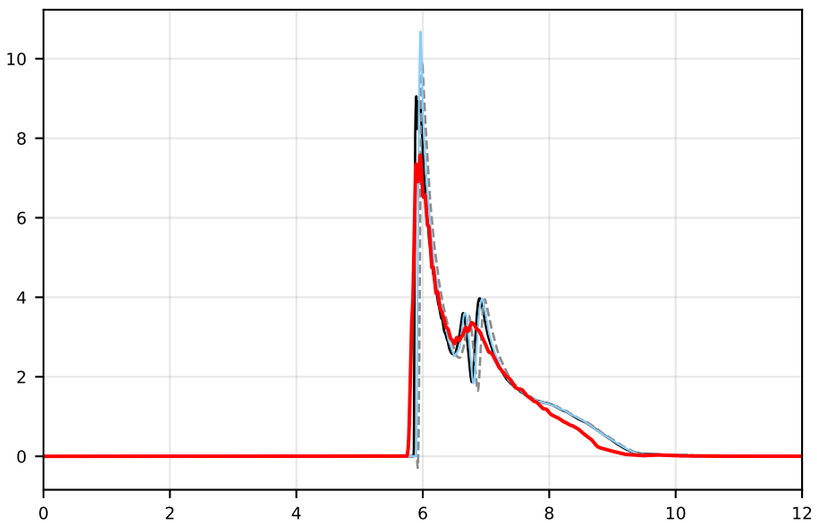
SeisSol O5

MFEM mixed-flux RK45 p=2 (gold)

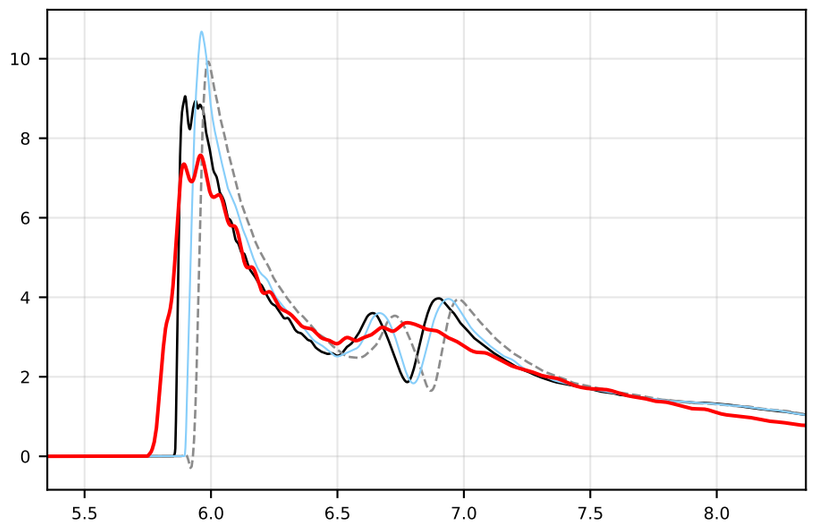
Nitsche p=1 200 m (Vista 1075919)

SCEC TPV104 station x2_-12_x3_3 (strike -12 km, depth 3 km)
Nitsche p=1 200 m (Vista 1075919) vs DRDG3D: dt -0.107 s slip -3.45% peak -16.7%

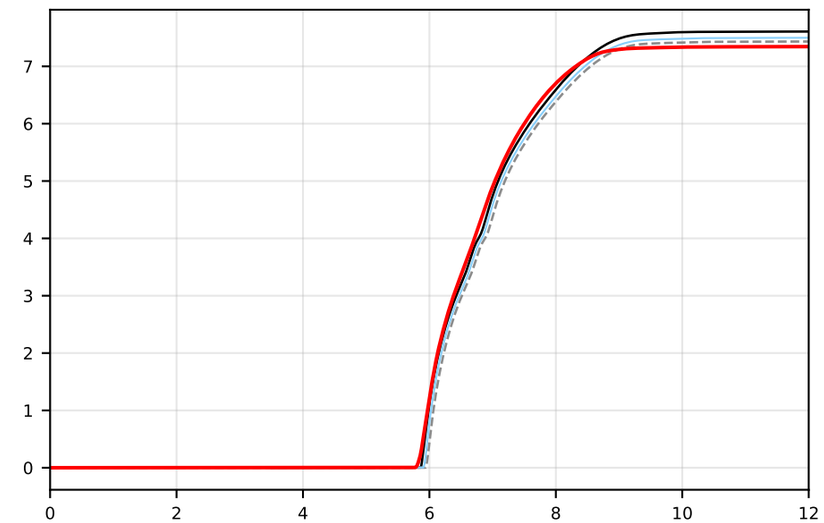
(a) strike slip rate (m/s)



(b) strike slip rate (m/s), DRDG3D arrival 5.853 s -0.5 / +2.5 s

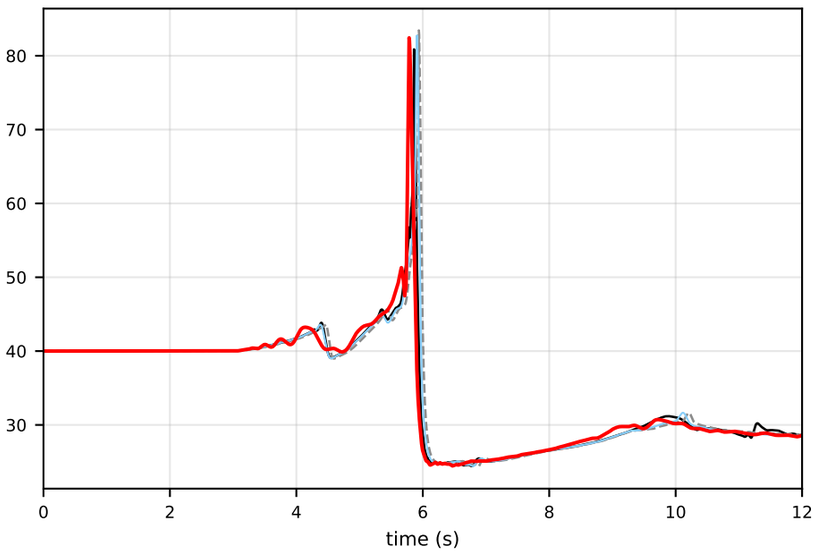


(c) strike slip (m)

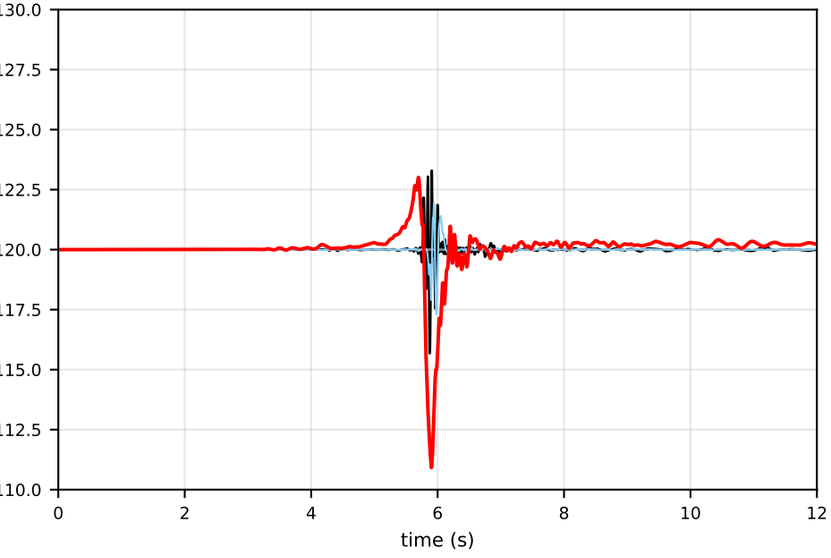


time (s)

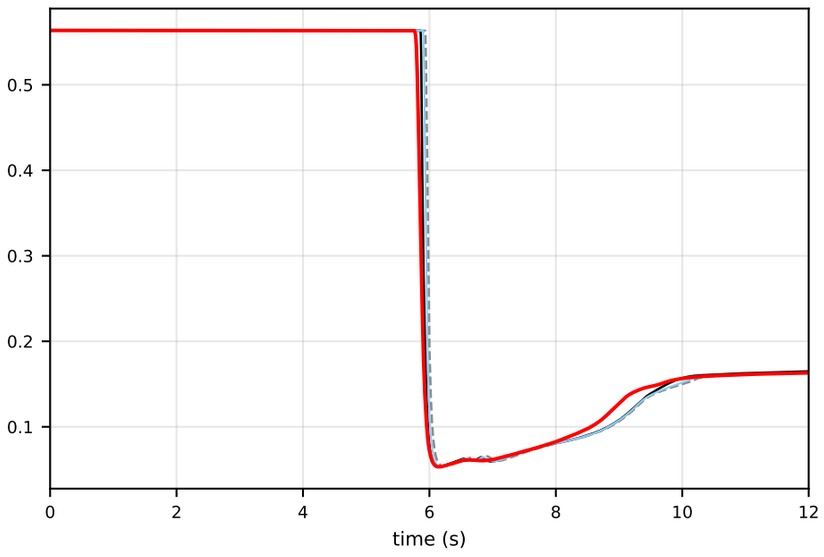
(d) strike shear stress (MPa)



(e) normal stress, compression positive (MPa); axis 120 +- 10 MPa

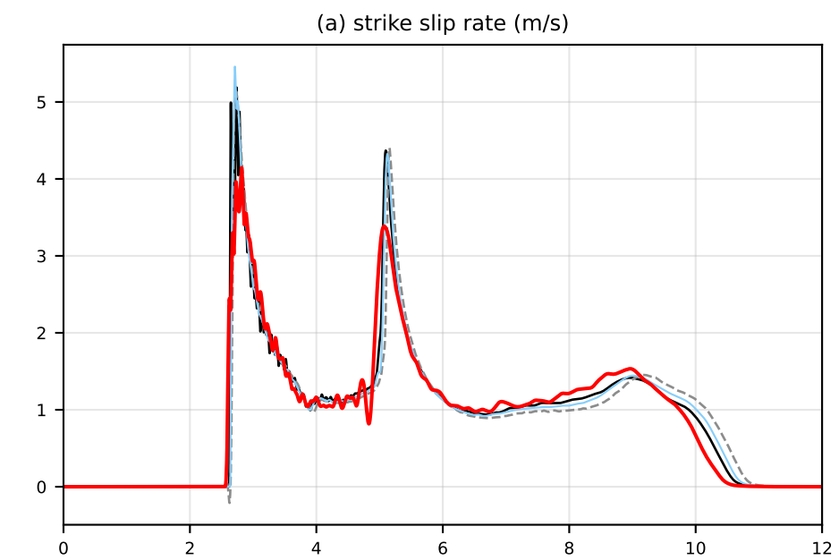


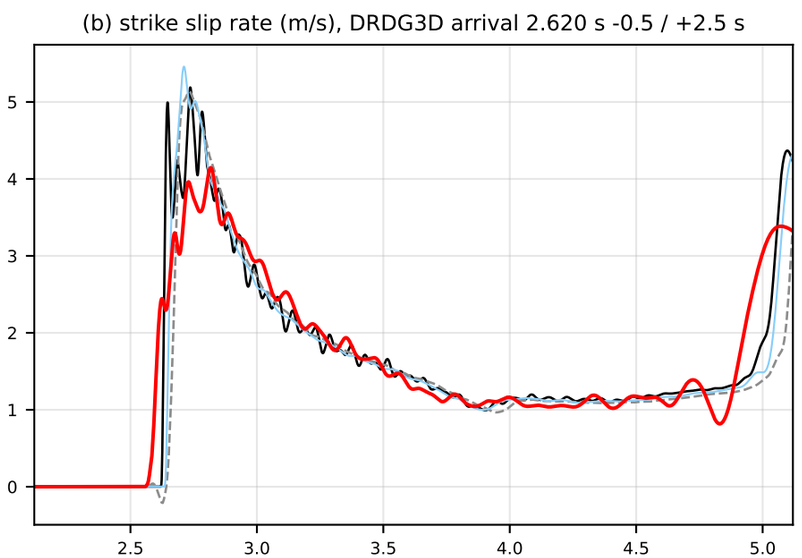
(f) state variable psi

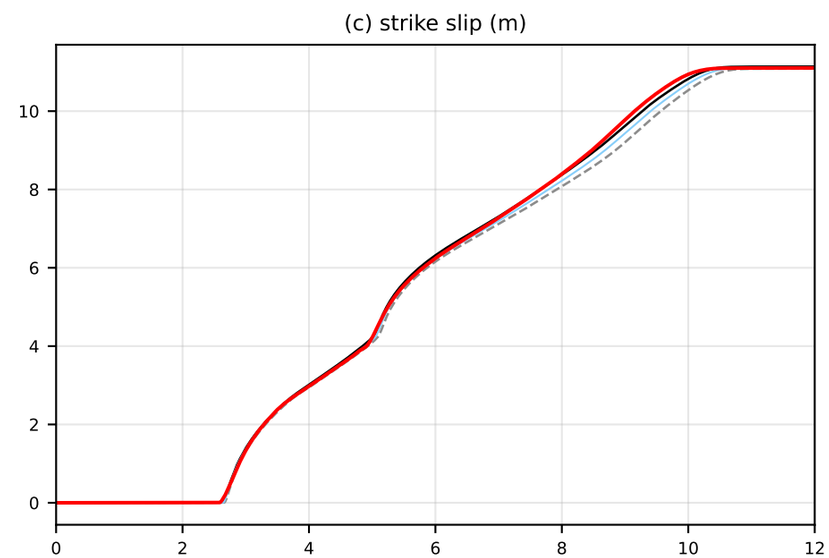


DRDG3D 200 m O4 | SeisSol O5 | MFEM mixed-flux RK45 p=2 (gold) | Nitsche p=1 200 m (Vista 1075919)

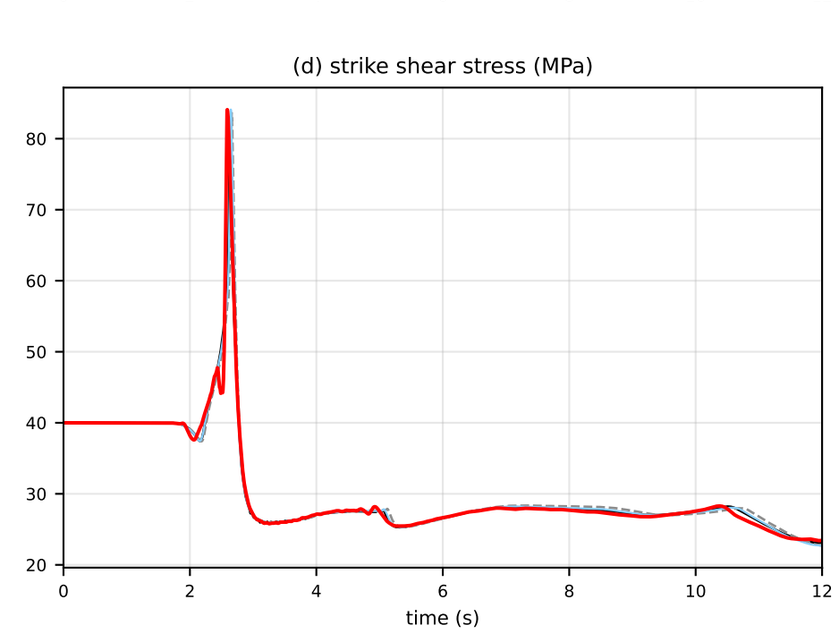
SCEC TPV104 station x2_0_x3_3 (strike 0 km, depth 3 km)
Nitsche p=1 200 m (Vista 1075919) vs DRDG3D: dt -0.060 s slip -0.32% peak -20.1%

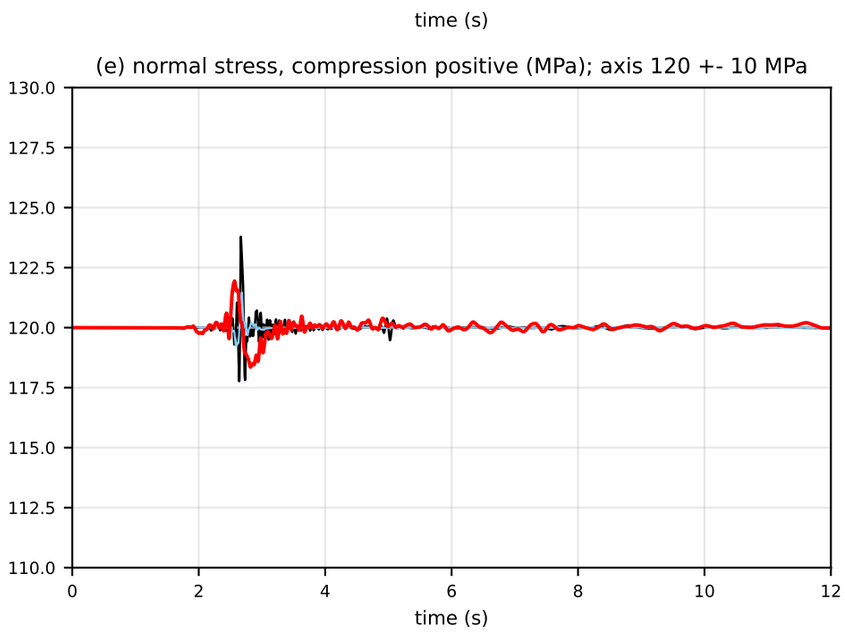


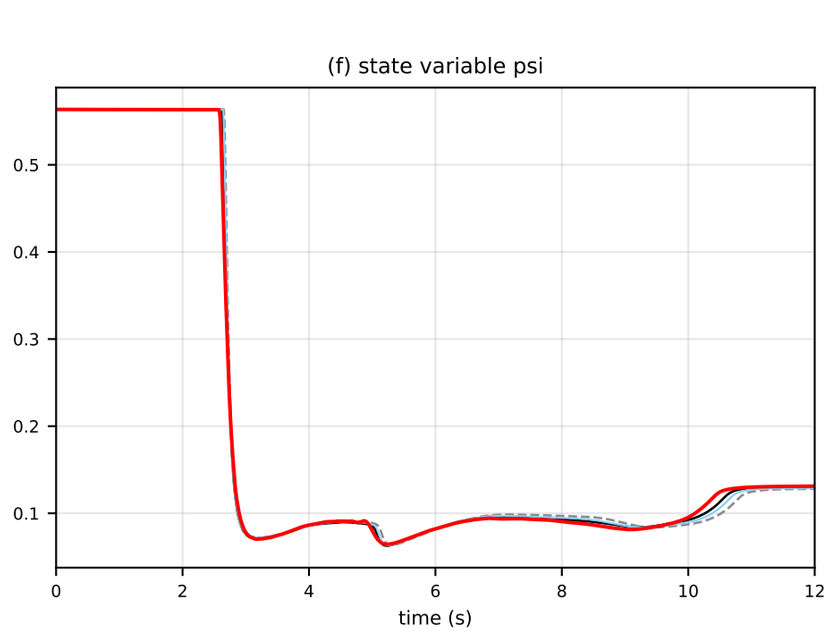




time (s)







DRDG3D 200 m O4 | SeisSol O5 | MFEM mixed-flux RK45 p=2 (gold) | Nitsche p=1 200 m (Vista 1075919)

SCEC TPV104 station x2_12_x3_3 (strike 12 km, depth 3 km)
Nitsche p=1 200 m (Vista 1075919) vs DRDG3D: dt -0.088 s slip -3.46% peak -16.3%

(a) strike slip rate (m/s)

(b) strike slip rate (m/s), DRDG3D arrival 5.854 s -0.5 / +2.5 s

(c) strike slip (m)

time (s)

(d) strike shear stress (MPa)

(e) normal stress, compression positive (MPa); axis 120 +- 10 MPa

(f) state variable psi

time (s)

time (s)

DRDG3D 200 m O4
SeisSol O5
MFEM mixed-flux RK45 p=2 (gold)
Nitsche p=1 200 m (Vista 1075919)

SCEC TPV104 station x2_-9_x3_7.5 (strike -9 km, depth 7.5 km)
Nitsche p=1 200 m (Vista 1075919) vs DRDG3D: dt -0.162 s slip -0.49% peak -22.7%

(a) strike slip rate (m/s)

(b) strike slip rate (m/s), DRDG3D arrival 4.229 s -0.5 / +2.5 s

(c) strike slip (m)

time (s)

(d) strike shear stress (MPa)

(e) normal stress, compression positive (MPa); axis 120 +- 10 MPa

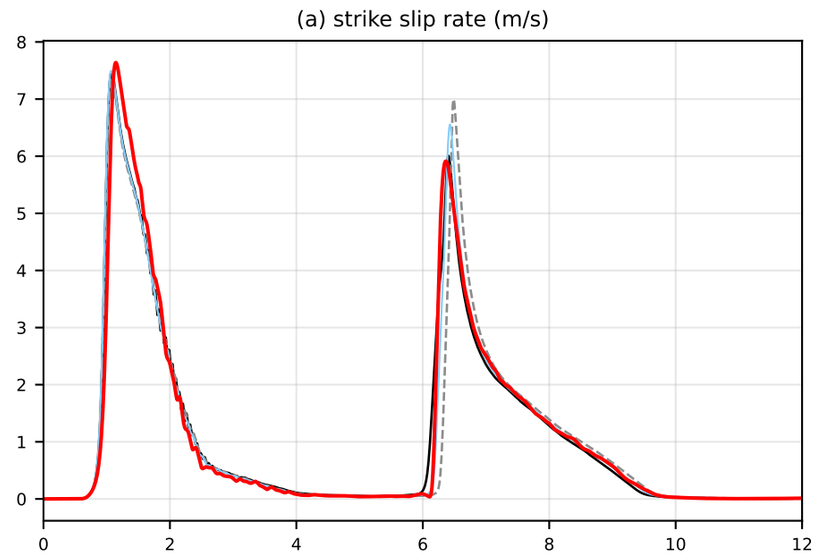
(f) state variable psi

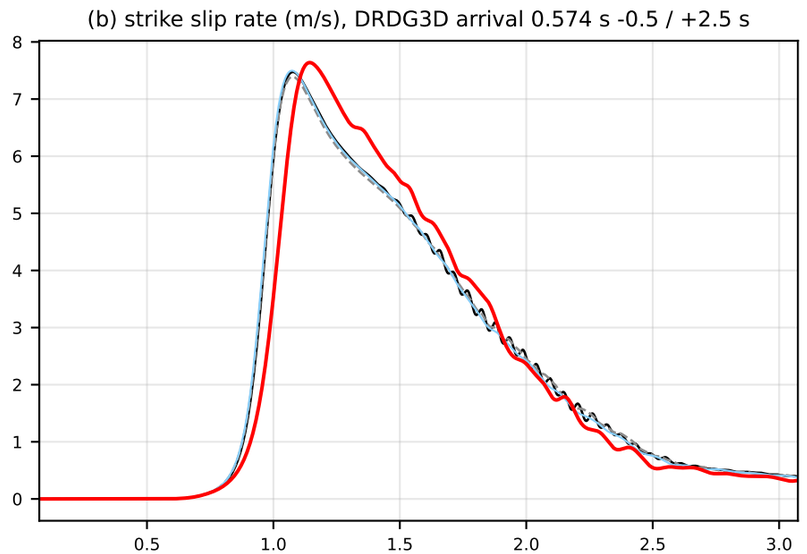
time (s)

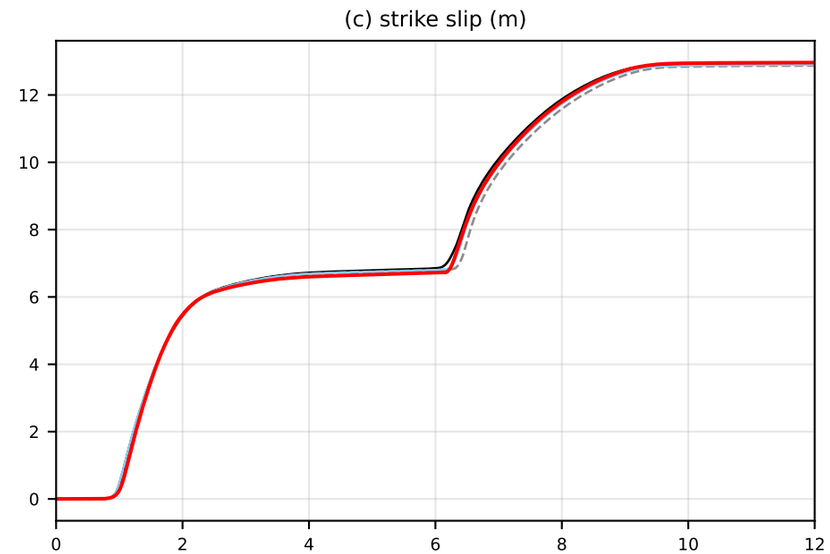
time (s)

DRDG3D 200 m O4 | SeisSol O5 | MFEM mixed-flux RK45 p=2 (gold) | Nitsche p=1 200 m (Vista 1075919)

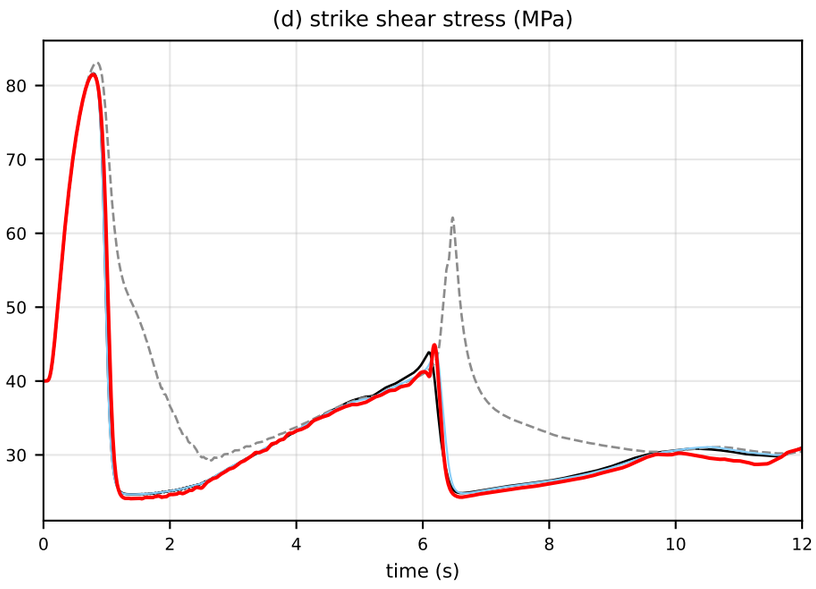
SCEC TPV104 station x2_0_x3_7.5 (strike 0 km, depth 7.5 km)
Nitsche p=1 200 m (Vista 1075919) vs DRDG3D: dt -0.003 s slip +0.25% peak +2.2%

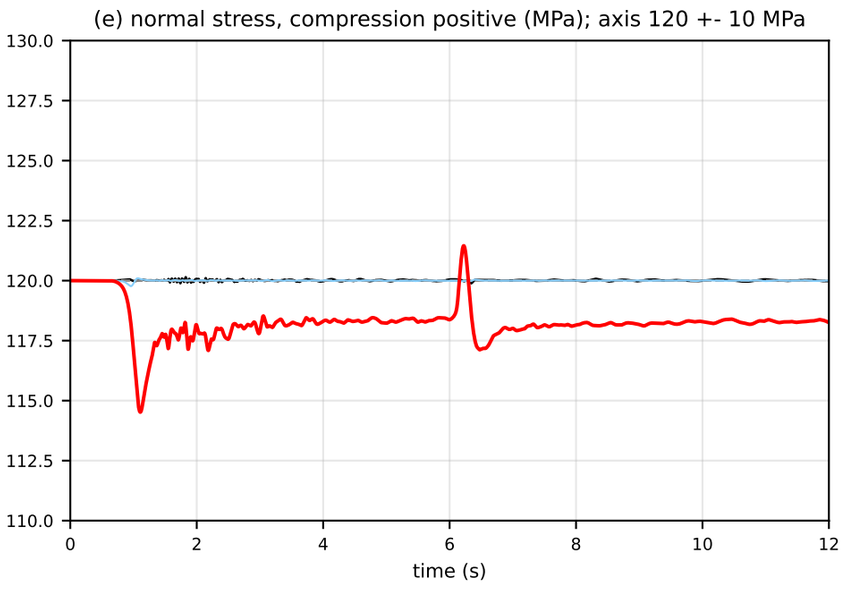


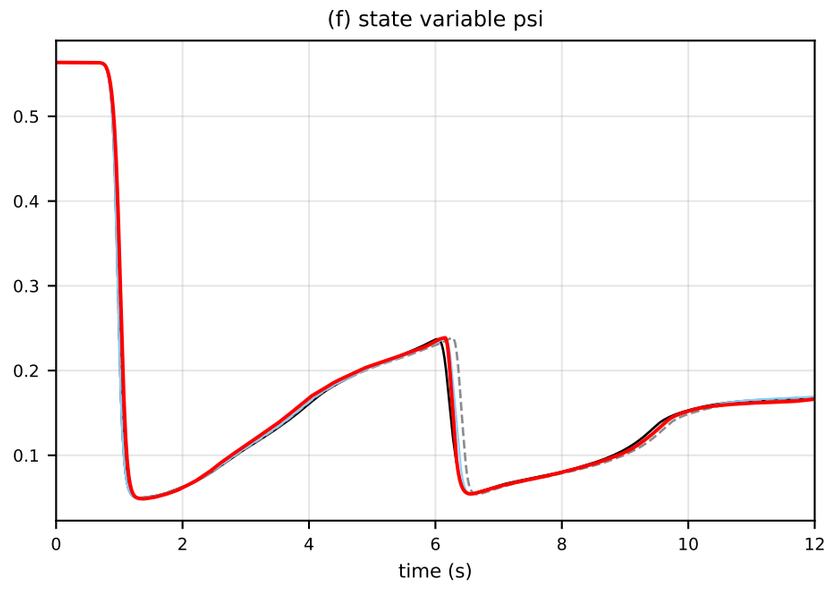




time (s)

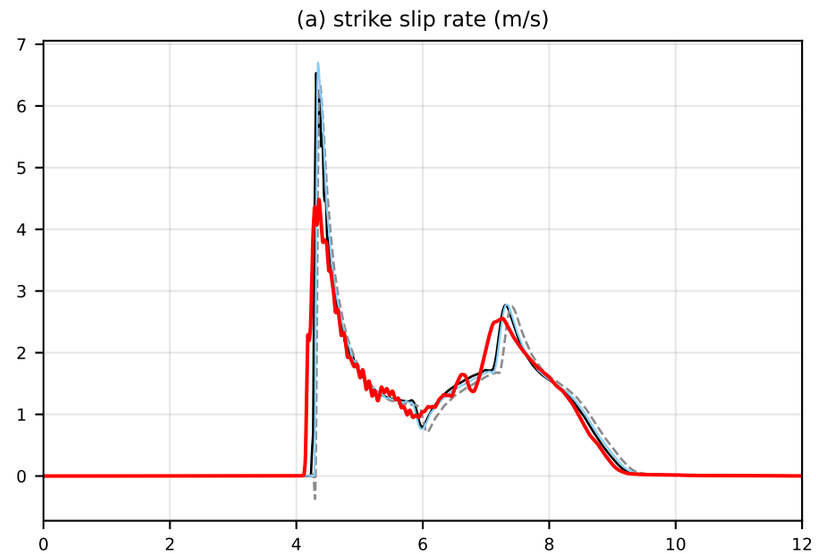


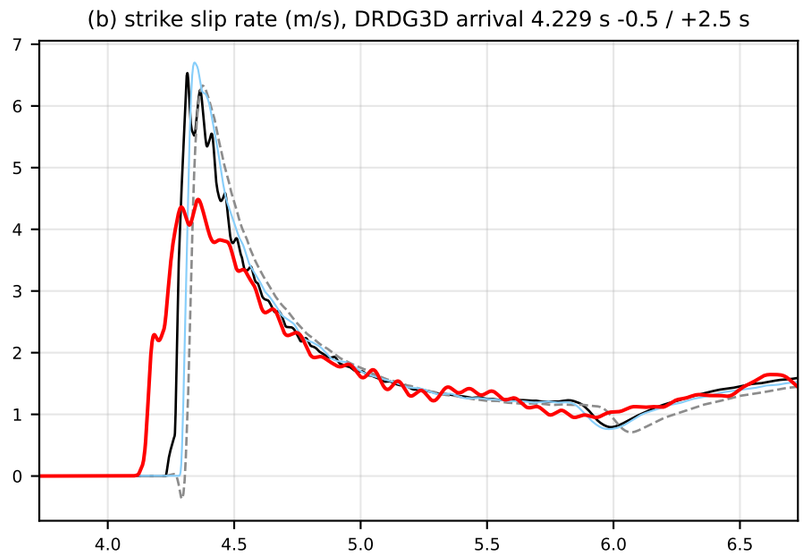


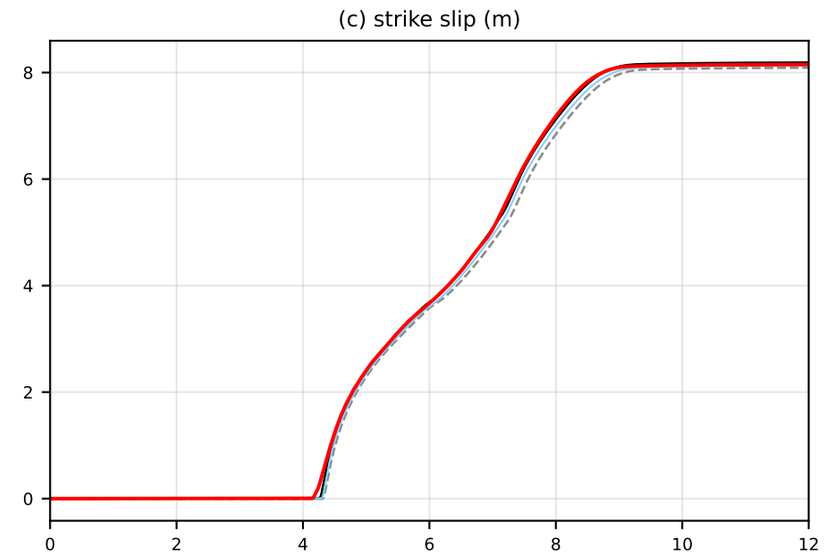


DRDG3D 200 m O4 | SeisSol O5 | MFEM mixed-flux RK45 p=2 (gold) | Nitsche p=1 200 m (Vista 1075919)

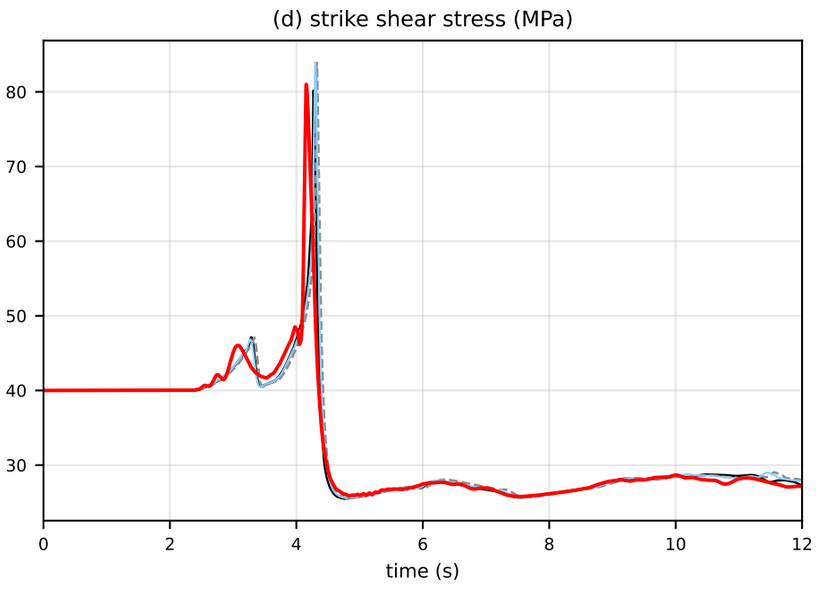
SCEC TPV104 station x2_9_x3_7.5 (strike 9 km, depth 7.5 km)
Nitsche p=1 200 m (Vista 1075919) vs DRDG3D: dt -0.131 s slip -0.46% peak -31.4%

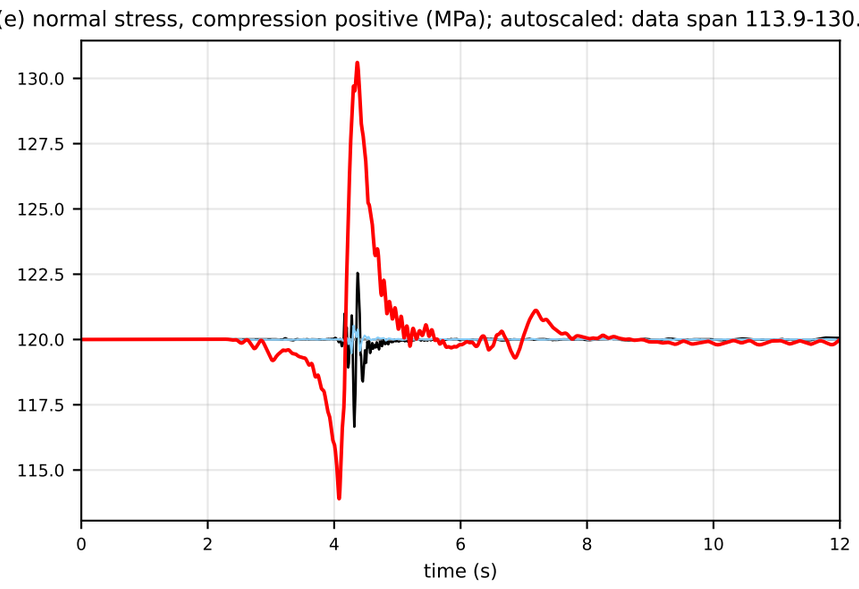


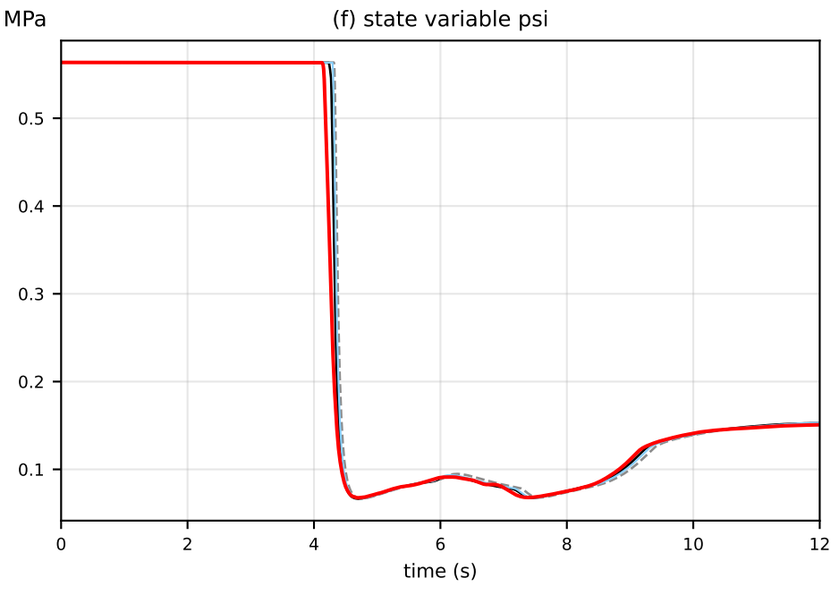




time (s)



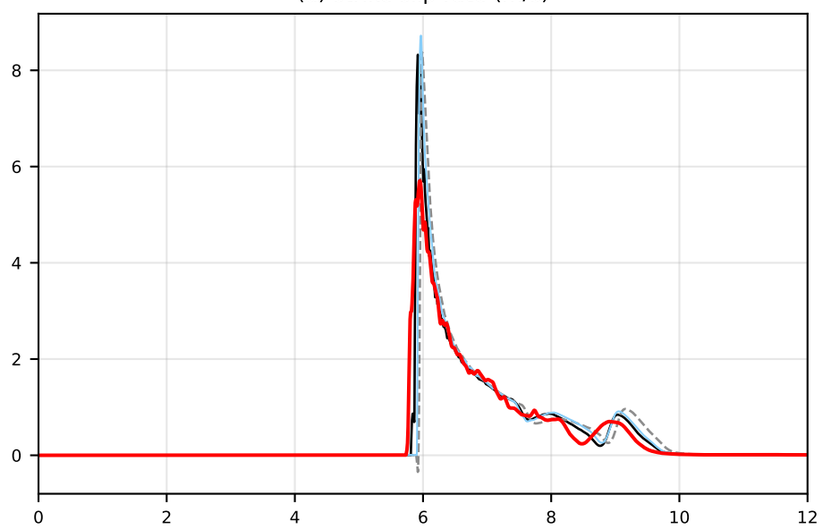




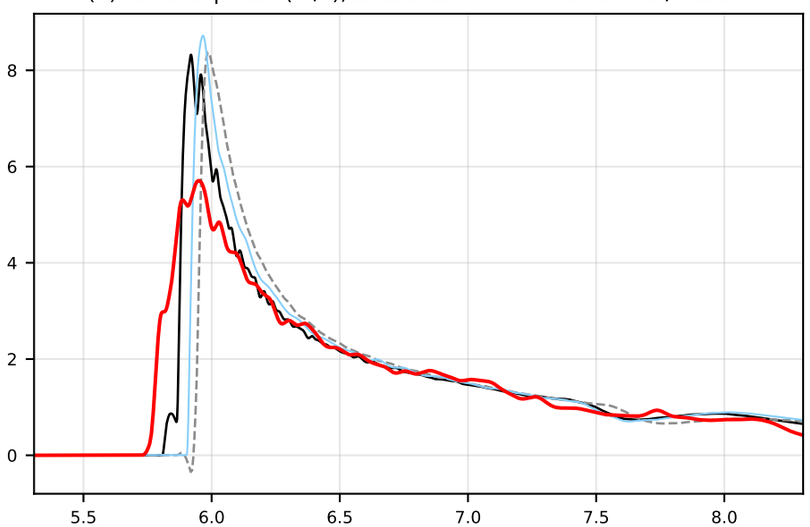
DRDG3D 200 m O4 | SeisSol O5 | MFEM mixed-flux RK45 p=2 (gold) | Nitsche p=1 200 m (Vista 1075919)

SCEC TPV104 station x2_-12_x3_12 (strike -12 km, depth 12 km)
Nitsche p=1 200 m (Vista 1075919) vs DRDG3D: dt -0.079 s slip -2.67% peak -31.5%

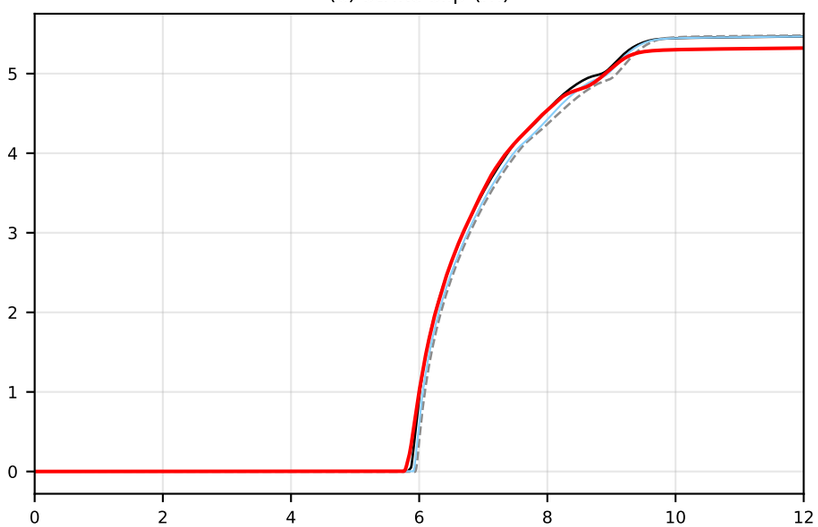
(a) strike slip rate (m/s)



(b) strike slip rate (m/s), DRDG3D arrival 5.807 s -0.5 / +2.5 s

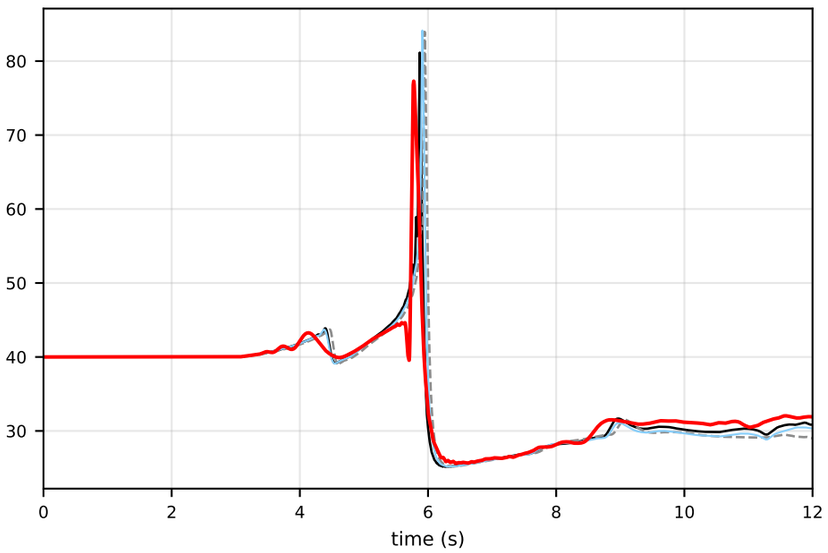


(c) strike slip (m)

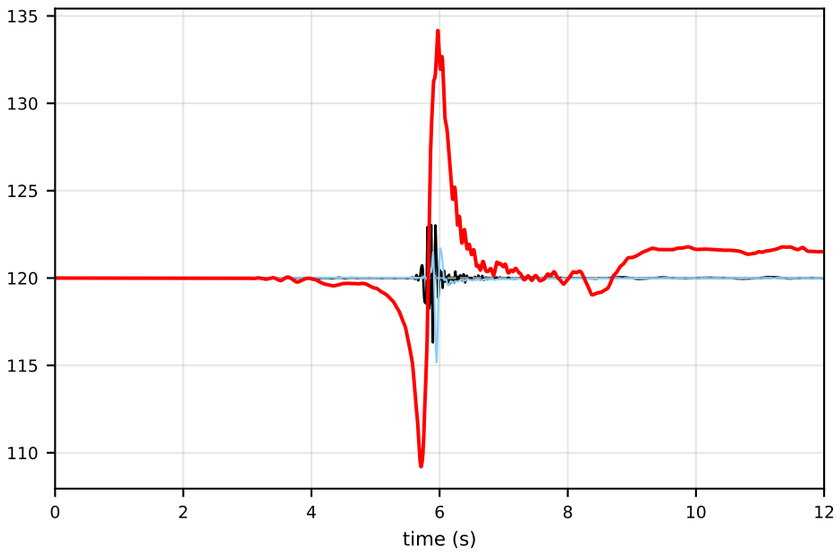


time (s)

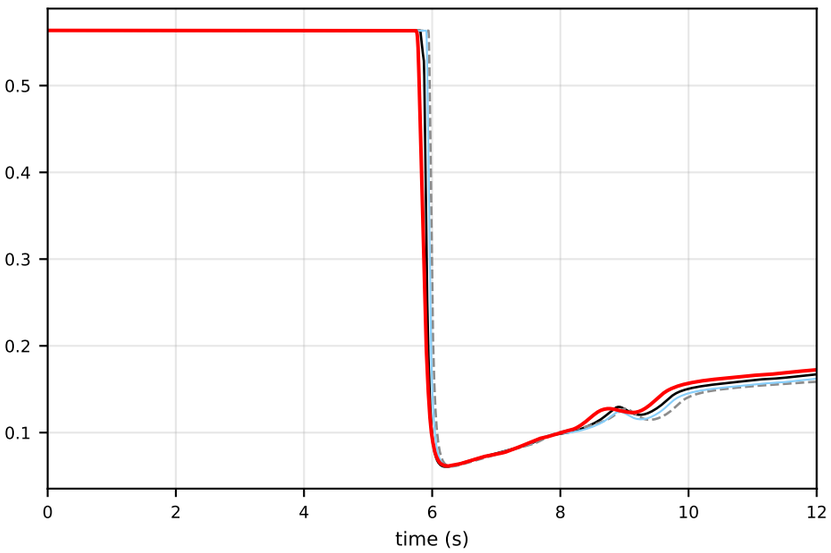
(d) strike shear stress (MPa)



(e) normal stress, compression positive (MPa); autoscaled: data span 109.2-134.2 MPa

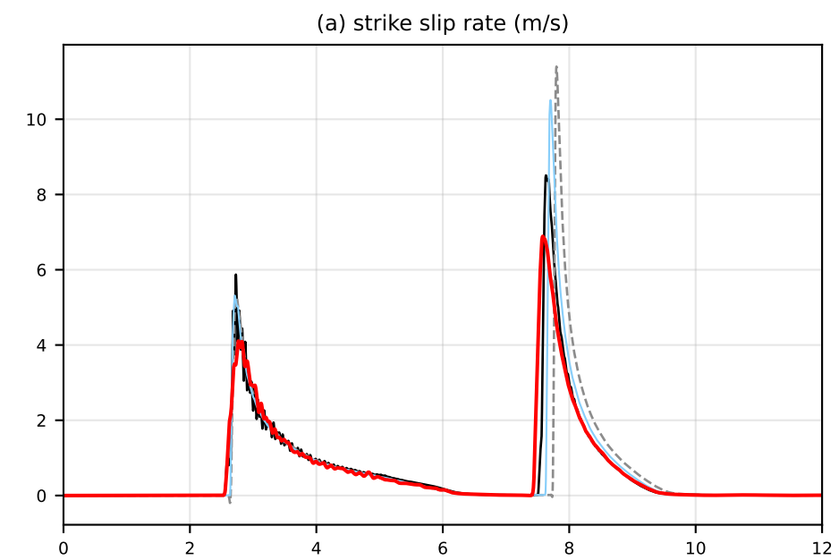


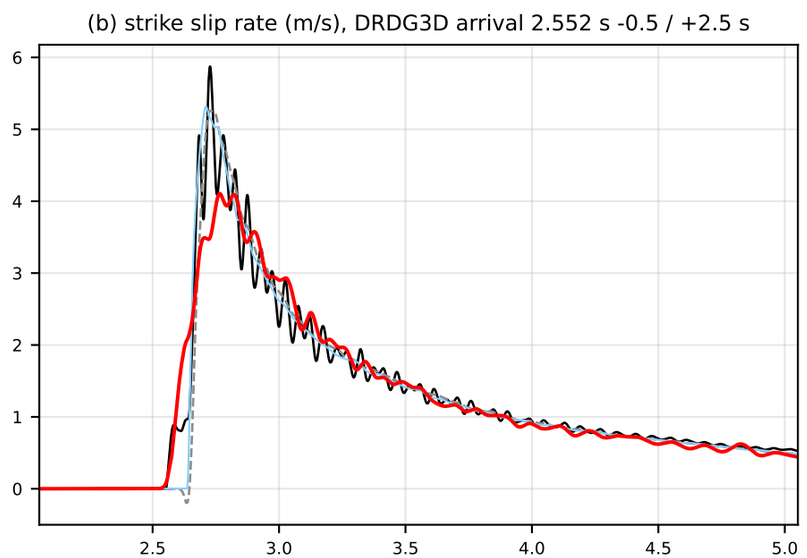
(f) state variable psi

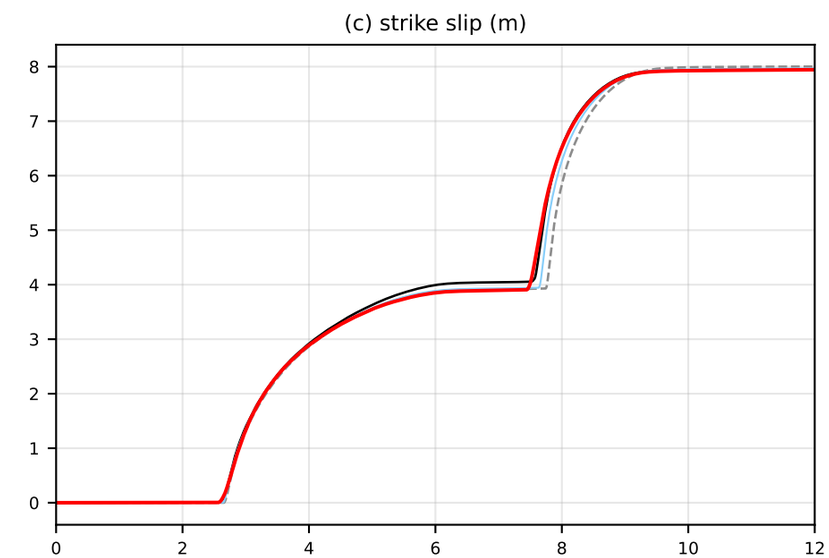


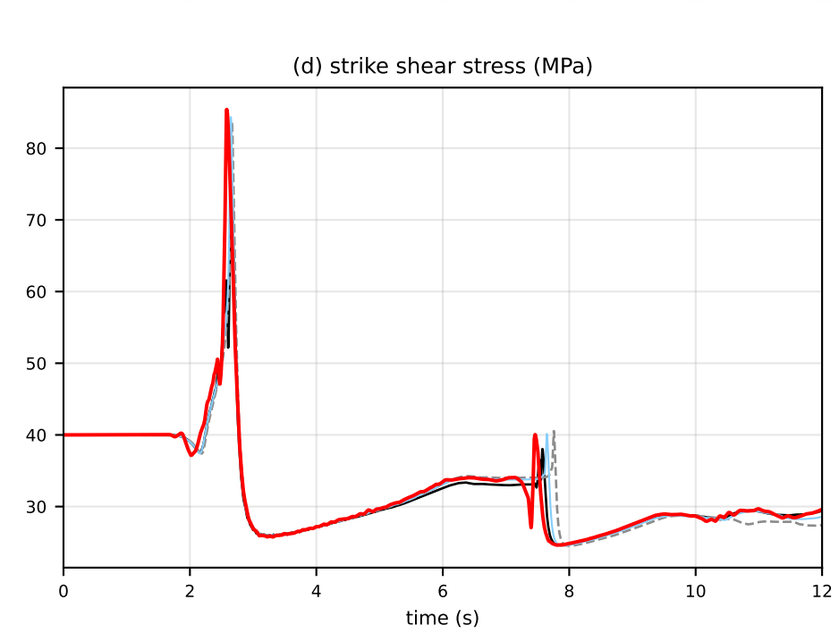
DRDG3D 200 m O4
SeisSol O5
MFEM mixed-flux RK45 p=2 (gold)
Nitsche p=1 200 m (Vista 1075919)

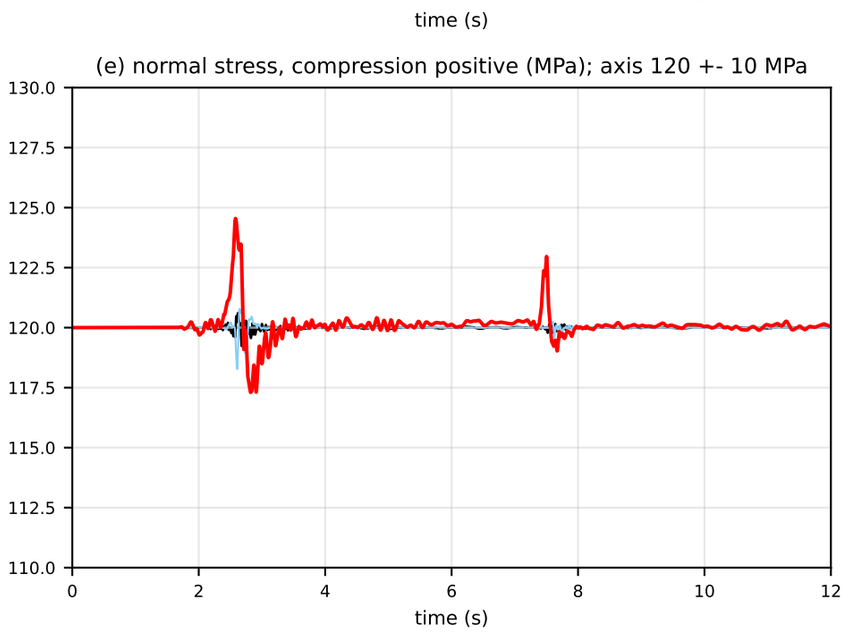
SCEC TPV104 station x2_0_x3_12 (strike 0 km, depth 12 km)
Nitsche p=1 200 m (Vista 1075919) vs DRDG3D: dt -0.026 s slip -0.11% peak -19.0%

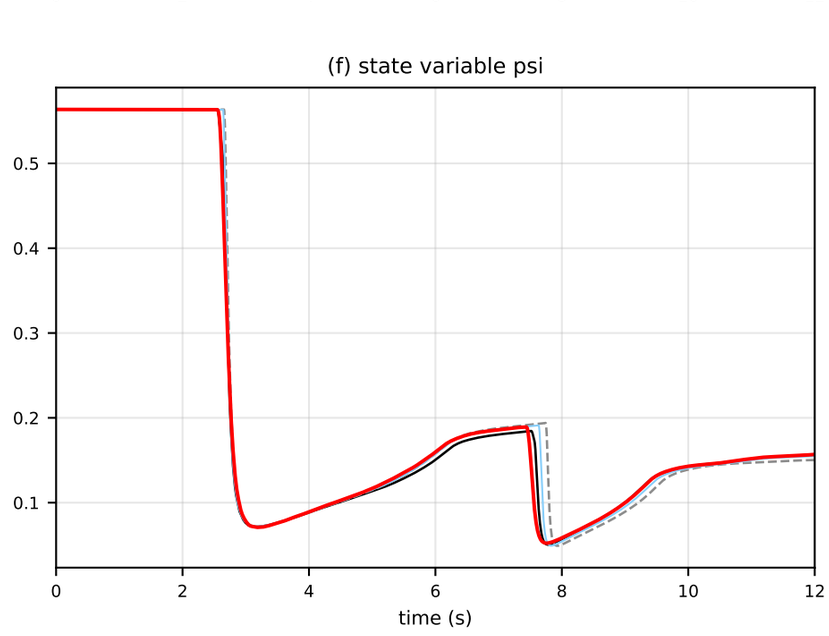












DRDG3D 200 m O4 | SeisSol O5 | MFEM mixed-flux RK45 p=2 (gold) | Nitsche p=1 200 m (Vista 1075919)

SCEC TPV104 station x2_12_x3_12 (strike 12 km, depth 12 km)
Nitsche p=1 200 m (Vista 1075919) vs DRDG3D: dt -0.062 s slip -2.58% peak -33.8%

(a) strike slip rate (m/s)

(b) strike slip rate (m/s), DRDG3D arrival 5.809 s -0.5 / +2.5 s

(c) strike slip (m)

time (s)

(d) strike shear stress (MPa)

(e) normal stress, compression positive (MPa); autoscaled: data span 106.0-128.4 MPa

(f) state variable psi

time (s)

time (s)

DRDG3D 200 m O4
SeisSol O5
MFEM mixed-flux RK45 p=2 (gold)
Nitsche p=1 200 m (Vista 1075919)